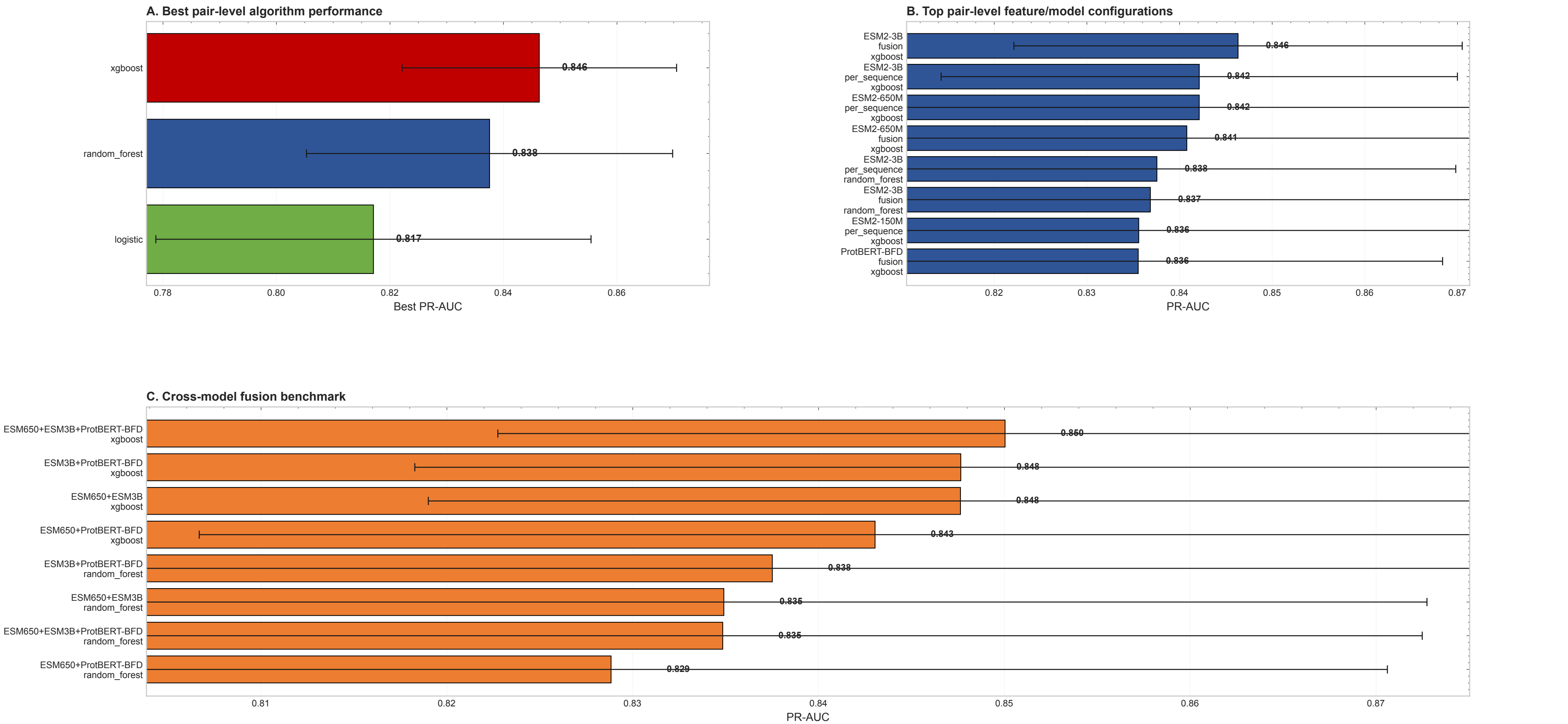


Figure 4. Pair-level machine learning and cross-model fusion benchmarks



D. Top classical and fusion models						
Group	Model	ROC-AUC	PR-AUC	F1	Accuracy	Brier
Cross-model fusion	ESM650+ESM3B+ProtBERT-BFD xgboost	0.766	0.850	0.768	0.700	0.212
Cross-model fusion	ESM3B+ProtBERT-BFD xgboost	0.762	0.848	0.759	0.690	0.216
Cross-model fusion	ESM650+ESM3B xgboost	0.764	0.848	0.763	0.693	0.212
Pair-level ML	ESM2-3B fusion xgboost	0.759	0.846	0.756	0.688	0.215
Cross-model fusion	ESM650+ProtBERT-BFD xgboost	0.757	0.843	0.761	0.691	0.217
Pair-level ML	ESM2-3B per_sequence xgboost	0.757	0.842	0.761	0.691	0.217
Pair-level ML	ESM2-650M per_sequence xgboost	0.758	0.842	0.763	0.694	0.215
Pair-level ML	ESM2-650M fusion xgboost	0.756	0.841	0.760	0.692	0.216
Pair-level ML	ESM2-3B per_sequence random_forest	0.754	0.838	0.773	0.671	0.204
Cross-model fusion	ESM3B+ProtBERT-BFD random_forest	0.754	0.838	0.771	0.671	0.203